Bright Learn Data Analytics Coding

Practical 1: SQL Fundamentals (DataBricks-Basic SQL Syntax)

Data Science Group

```sql
---1. SELECT Statement
-----------------------------------------------------------------------------
--- Q1. Display all columns for all transactions
select * from `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table +

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
3	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
4	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
5	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
6	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
7	7	2023-03-13	CUST007	Male	46	Clothing	2	25	50
8	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
9	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
10	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
11	11	2023-02-14	CUST011	Male	23	Clothing	2	50	100
12	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
13	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
14	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
15	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
16	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500

1,000 rows | 0.85s runtime Refreshed 2 minutes ago

```sql
--- Q2. Display only the Transaction ID, Da Customer ID for all records.
Select 'Transaction ID', Date, 'Customer ID'
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table +

	Transaction ID	Date	Customer ID
1	Transaction ID	2023-11-24	Customer ID
2	Transaction ID	2023-02-27	Customer ID
3	Transaction ID	2023-01-13	Customer ID
4	Transaction ID	2023-05-21	Customer ID
5	Transaction ID	2023-05-06	Customer ID
6	Transaction ID	2023-04-25	Customer ID
7	Transaction ID	2023-03-13	Customer ID
8	Transaction ID	2023-02-22	Customer ID
9	Transaction ID	2023-12-13	Customer ID
10	Transaction ID	2023-10-07	Customer ID
11	Transaction ID	2023-02-14	Customer ID
12	Transaction ID	2023-10-30	Customer ID
13	Transaction ID	2023-08-05	Customer ID
14	Transaction ID	2023-01-17	Customer ID
15	Transaction ID	2023-01-16	Customer ID
16	Transaction ID	2023-02-17	Customer ID
17	Transaction ID	2023-04-22	Customer ID

1,000 rows | 1.10s runtime

```sql
--- Q3. Display all the distinct product categories in the dataset.
Select Distinct `Product Category`
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Product Category
1	Beauty
2	Clothing
3	Electronics

```sql
--- Q5. Display nsactions where the Age is greater than 40.
Select Distinct *
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
Where Age > 40;
```

Add parameter

Table

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
2	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
3	7	2023-03-13	CUST007	Male	46	Clothing	2	25	50
4	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
5	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
6	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
7	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
8	18	2023-04-30	CUST018	Female	47	Electronics	2	25	50
9	19	2023-09-16	CUST019	Female	62	Clothing	2	25	50
10	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
11	24	2023-11-29	CUST024	Female	49	Clothing	1	300	300
12	25	2023-12-26	CUST025	Female	64	Beauty	1	50	50
13	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
14	29	2023-08-18	CUST029	Female	42	Electronics	1	30	30
15	31	2023-05-23	CUST031	Male	44	Electronics	4	300	1200
16	33	2023-03-23	CUST033	Female	50	Electronics	2	50	100

534 rows | 0.68s runtime

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```sql
--- Q6. Display all transactions where the Price per Unit is between 100 and 500. Expected output: All columns
Select *
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
Where `Price per Unit` between 100 AND 500;
```

Add parameter

Table

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
3	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
4	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
5	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
6	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500
7	20	2023-11-05	CUST020	Male	22	Clothing	3	300	900
8	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
9	24	2023-11-29	CUST024	Female	49	Clothing	1	300	300
10	26	2023-10-07	CUST026	Female	28	Electronics	2	500	1000
11	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
12	30	2023-10-29	CUST030	Female	39	Beauty	3	300	900
13	31	2023-05-23	CUST031	Male	44	Electronics	4	300	1200
14	35	2023-08-05	CUST035	Female	58	Beauty	3	300	900
15	36	2023-06-24	CUST036	Male	52	Beauty	3	300	900
16	42	2023-02-17	CUST042	Male	22	Clothing	3	300	900

396 rows | 0.69s runtime

Refreshed now

```sql
--- Q7. Display all transactions where the Product Category is either 'Beauty' or 'Electronics'
Select *
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
Where`Product Category` IN ('Beauty','Electronics') ;
```

Add parameter

Table

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
4	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
5	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
6	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
7	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
8	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
9	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
10	18	2023-04-30	CUST018	Female	47	Electronics	2	25	50
11	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
12	25	2023-12-26	CUST025	Female	64	Beauty	1	50	50
13	26	2023-10-07	CUST026	Female	28	Electronics	2	500	1000
14	27	2023-08-03	CUST027	Female	38	Beauty	2	25	50
15	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
16	29	2023-08-18	CUST029	Female	42	Electronics	1	30	30

649 rows | 0.65s runtime

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```sql
--- Q8. Display all transactions where the Product Category is not 'Clothing'.
Select *
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
Where `Product Category` != 'Clothing';
```

Add parameter

Table

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
4	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
5	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
6	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
7	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
8	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
9	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
10	18	2023-04-30	CUST018	Female	47	Electronics	2	25	50
11	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
12	25	2023-12-26	CUST025	Female	64	Beauty	1	50	50
13	26	2023-10-07	CUST026	Female	28	Electronics	2	500	1000
14	27	2023-08-03	CUST027	Female	38	Beauty	2	25	50
15	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
16	29	2023-08-18	CUST029	Female	42	Electronics	1	30	30
17	30	2023-10-29	CUST030	Female	39	Beauty	3	300	900

649 rows | 0.42s runtime

Refreshed now

```sql
--- Q9. Display all transactions where the Quantity is greater than or equal to 3.
Select *
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
Where `Quantity` >= 3;
--------------------------------------------------------------------------------
```

Add parameter

Table

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
3	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
4	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
5	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
6	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
7	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
8	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500
9	17	2023-04-22	CUST017	Female	27	Clothing	4	25	100
10	20	2023-11-05	CUST020	Male	22	Clothing	3	300	900
11	23	2023-04-12	CUST023	Female	35	Clothing	4	30	120
12	30	2023-10-29	CUST030	Female	39	Beauty	3	300	900
13	31	2023-05-23	CUST031	Male	44	Electronics	4	300	1200
14	32	2023-01-04	CUST032	Male	30	Beauty	3	30	90
15	34	2023-12-24	CUST034	Female	51	Clothing	3	50	150
16	35	2023-08-05	CUST035	Female	58	Beauty	3	300	900
17	36	2023-06-24	CUST036	Male	52	Beauty	3	300	900

504 rows | 0.28s runtime

Refreshed now

```sql
--- Q10. Count the total number of transactions.
Select Count(*) as Total_Transactions
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;

--- Q11. Find the average Age of customers. Expected output: Average Age
```

Add parameter

Table

	Total_Transactions
1	1000

```sql
--- Q11. Find the average Age of customers. Expected output:
Select AVG(`Age`) AS Average_Age
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Average_Age
1	41.392

```sql
--- Q12. Find the total quantity of products sold.
Select SUM(`Quantity`) AS Total_Quantity
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Total_Quantity
1	2514

```sql
--- Q13. Find the maximum Total Amount spent in a single transaction.
Select MAX(`Total Amount`) AS Max_Total_Amount
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Max_Total_Am...
1	2000

```sql
--- Q14. Find the minimum Price per Unit in the dataset. Expected output:
Select MIN(`Price per Unit`) AS Min_Price_per_Unit
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Min_Price_per_Unit
1	25

```sql
--- Q15. Find the number of transactions per Product Category. Expected output:
Select `Product Category`, COUNT(*) AS Transaction_Count
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
GROUP BY `Product Category`;
```

Add parameter

Table

	Product Category	Transaction_Count
1	Beauty	307
2	Clothing	351
3	Electronics	342

```sql
--- Q16. Find the total revenue (Total Amount) per gender.
Select `Gender`, SUM(`Total Amount`) AS Total_Revenue
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
GROUP BY `Gender`;
```

Add parameter

Table

	Gender	Total_Revenue
1	Male	223160
2	Female	232840

```sql
--- Q17. Find the average Price per Unit per product category.
Select `Product Category`, AVG(`Price per Unit`) AS Average_Price
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
GROUP BY `Product Category`;
```

Add parameter

Table

	Product Category	Average_Price
1	Beauty	184.05537459283389
2	Clothing	174.28774928774928
3	Electronics	181.90058479532163

```sql
--- Q18. Find the total revenue per product category where total revenue is greater than 10,000.
Select `Product Category`, SUM(`Price per Unit`) AS Total_Revenue
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
GROUP BY `Product Category`
HAVING  SUM(`Price per Unit`)> 10000;
```

Add parameter

Table

	Product Category	Total_Revenue
1	Beauty	56505
2	Clothing	61175
3	Electronics	62210

```sql
--- Q19. Find the average quantity per product category where the average is more than 2
Select `Product Category`, AVG(`Quantity`) AS Average_Quantity
From `workspace`.`default`.`1772863649520_retail_sales_dataset`
GROUP BY `Product Category`
HAVING AVG(`Quantity`) > 2;
```

Add parameter

Table

	Product Category	Average_Quantity
1	Beauty	2.511400651465798
2	Clothing	2.547008547008547
3	Electronics	2.482456140350877

```sql
--- Q20. Display a column called Spending_Level that shows 'High' if Total Amount > 1000, otherwise 'Low'. Spending_Level
Select `Transaction ID`,`Total Amount`,
CASE
   WHEN `Total Amount`> 1000 THEN 'High'
   ELSE 'Low'
END AS Spending_Level
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Transaction ID	Total Amount	Spending_Level
9	9	600	Low
10	10	200	Low
11	11	100	Low
12	12	75	Low
13	13	1500	High
14	14	120	Low
15	15	2000	High
16	16	1500	High
17	17	100	Low
18	18	50	Low
19	19	50	Low

1,000 rows | 1.76s runtime

```sql
--- Q21. Display a new column called Age_Group that labels customers as:
Select `Customer ID`,`Age`,
CASE
   WHEN `Age`>= 60 THEN 'Senior'
   WHEN `Age` BETWEEN 30 AND 59 THEN 'Adult'
   ELSE 'Youth'
END AS Age_Group
From `workspace`.`default`.`1772863649520_retail_sales_dataset`;
```

Add parameter

Table

	Customer ID	Age	Age_Group
1	CUST001	34	Adult
2	CUST002	26	Youth
3	CUST003	50	Adult
4	CUST004	37	Adult
5	CUST005	30	Adult
6	CUST006	45	Adult
7	CUST007	46	Adult
8	CUST008	30	Adult
9	CUST009	63	Senior
10	CUST010	52	Adult
11	CUST011	23	Youth

1,000 rows | 0.84s runtime